



ML-model ON crop classification and recommendation system using satellite data on Indian agriculture

B.Tech Project Report (2025-2026)

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Abstract

Agriculture in India faces increasing challenges due to climate variability, resource constraints, and the need for timely, data-driven decision making. This project presents an integrated crop classification and crop recommendation system developed for the state of Uttar Pradesh using satellite remote sensing, machine learning, and deep learning techniques. Multi-temporal Sentinel-2 imagery was utilized to classify approximately fifteen major crop types using supervised learning models such as Random Forest and Support Vector Machine, with ground truth data manually collected from Google Earth and Google Earth Street View. The crop classification model achieved a training accuracy of 99.89

Historical meteorological data from Agra, Uttar Pradesh, spanning 2020 to 2026, were used to train a Long Short-Term Memory (LSTM) based Recurrent Neural Network for weather forecasting. The predicted weather variables were combined with spatial land suitability analysis and agronomic rule-based reasoning to recommend suitable crops under forecasted climatic conditions. The proposed framework provides actionable insights for farmers to improve crop selection and reduce climate-related risks, while offering policymakers a scalable tool for monitoring cropping patterns and supporting evidence-based agricultural planning. The scope of the project was extended to focus on crops exhibiting high price volatility in Minimum Support Price (MSP) dynamics, with particular emphasis on onion as a case study. The extended study was structured into three major components. First, an improved crop classification model was developed to enhance robustness and accuracy under varying environmental and spectral conditions. Second, a price prediction framework was designed using Long Short-Term Memory (LSTM) networks, trained on historical onion price data from 2020 to 2025, to forecast future market trends. Third, a detailed market analysis was conducted on Nashik mandi prices over the same period to investigate price stabilization mechanisms.

The analysis critically examined the existing government interventions, such as import-export regulations, used to control onion price fluctuations. Based on observed patterns, the study proposes an alternative approach centered on decentralized storage systems, enabling farmers to store produce post-harvest and strategically sell during peak price periods. This shift from reactive market intervention to proactive storage-based stabilization has the potential

to improve farmer income resilience and reduce market shocks. All extended analyses were specifically conducted for Nashik mandis in Maharashtra for the period 2020 to 2025, providing a focused, data-driven case study within the broader agricultural decision-support framework.

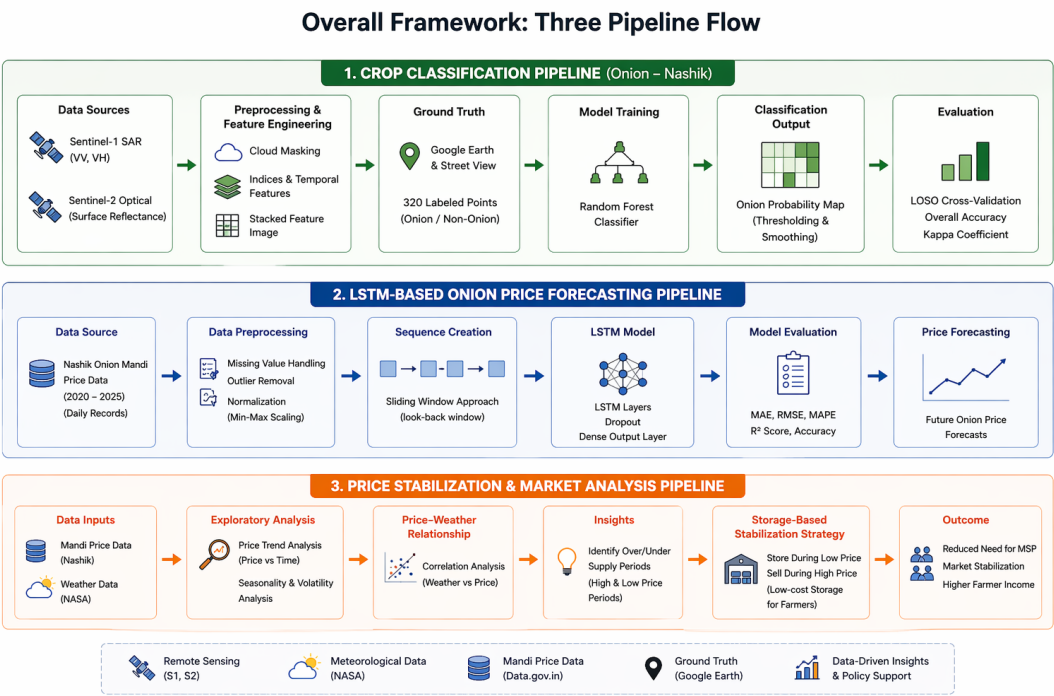


Figure 1: All 3 pipeline flow

1. Introduction

Agriculture today faces increasing uncertainty due to climate variability, land degradation, and inefficient resource utilization, making data-driven decision support systems essential. This project presents an integrated crop classification, crop recommendation, and market intelligence framework that leverages remote sensing, machine learning, and deep learning techniques to support informed agricultural decision-making.

The initial phase of the study focused on the state of Uttar Pradesh, selected for its agricultural diversity and significance. Multi-temporal satellite imagery from Sentinel-2, combined with meteorological data obtained from NASA, was used to capture crop dynamics and environmental conditions. Ground truth data were manually collected using Google Earth and Google Street View to ensure reliable crop labeling. Using these inputs, crop classification was performed across approximately fifteen major crop classes using supervised learning models such as Random Forests and Support Vector Machines (SVM), enabling accurate spatial mapping of cropping patterns. Following the mid-semester evaluation, the project scope was extended to address economic challenges associated with crop price volatility, with a focused case study on onion. An improved crop classification model was developed using enhanced feature engineering, multi-seasonal analysis, and additional radar-based inputs, with new ground truth data consisting of 320 manually labeled points collected from Nashik using Google Street View. This improved model strengthened classification robustness under diverse conditions and enabled more reliable crop identification.

In parallel, a price prediction model was developed using LSTM networks trained on historical onion price data from 2020 to 2025. Extensive exploratory data analysis (EDA) was conducted on Nashik mandi data, covering over 24,000 records across multiple markets, revealing strong seasonal trends, significant inter-annual variability, and pronounced price spikes during specific months. Additional analysis comparing producer markets in Nashik with consumer markets such as Mumbai showed a relatively small average price spread of approximately 75 per quintal, indicating that volatility is largely driven at the production level rather than along the supply chain.

Further, a detailed study of price-weather relationships highlighted moderate correlations between variables such as soil moisture, humidity, and temperature with onion prices, while precipitation showed minimal direct influence. These insights were incorporated into the forecasting framework to improve prediction reliability.

Based on the combined analysis, the study critically evaluates existing government price stabilization strategies, particularly import-export interventions, and proposes an alternative storage-based approach. By enabling farmers to store produce post-harvest and sell during peak price periods, this strategy has the potential to reduce market shocks and improve income stability. All extended analyses were conducted specifically for Nashik mandis in Maharashtra over the period 2020–2025, providing a focused, data-driven case study that complements the broader crop intelligence framework.

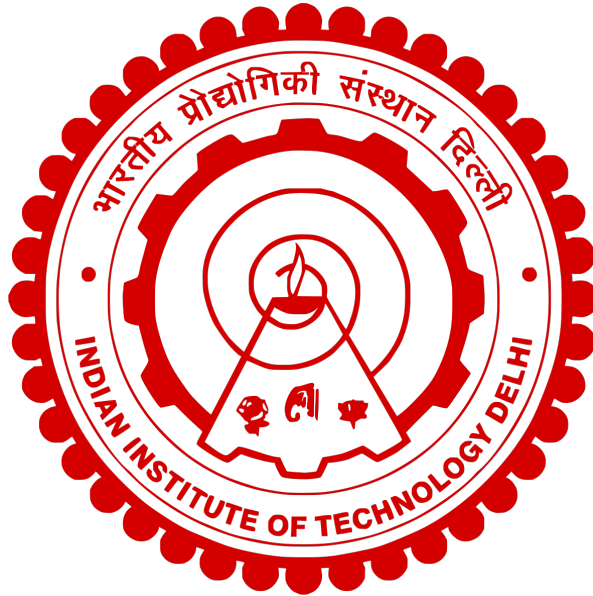


Figure 1.1

2. Methodology

2.1 Data Sources and Acquisition

This study integrates multi-source datasets spanning remote sensing, meteorology, and agricultural market data to support crop classification, recommendation, price prediction and price analysis.

For crop classification, multi-temporal satellite data from **Sentinel-1 (SAR)** and **Sentinel-2 (optical)** missions were utilized. Sentinel-1 Ground Range Detected (GRD) products with VV and VH polarizations (10m resolution) were used to capture structural and moisture-related crop characteristics under all-weather conditions. Complementary Sentinel-2 imagery provided spectral information for vegetation indices and phenological analysis. The satellite data span multiple agricultural seasons between **2022 and 2026** and were accessed via the Copernicus Open Access Hub.

Ground truth data were manually collected using **Google Earth and Google Street View** to ensure reliable labeling for supervised learning. For Uttar Pradesh, crop samples were annotated across multiple crop classes. In the extended study focused on **Nashik, Maharashtra**, an additional set of **320 ground truth points** was collected specifically for onion crop identification, improving model robustness for region-specific classification tasks.

Meteorological data were obtained from **NASA repositories** for two regions: Agra (Uttar Pradesh) and Nashik (Maharashtra), covering the period from **2020 to 2026**. The dataset includes key climatic variables such as temperature, relative humidity, precipitation, atmospheric pressure, wind speed, and soil moisture. These time-series data were used to train Long Short-Term Memory (LSTM)-based models for weather forecasting and to analyze climate-crop and climate-price relationships.

To study market dynamics, historical **mandi price data for onion** were collected from the **Data.gov.in** platform for Nashik district, spanning **2020 to 2025**. The dataset consists of daily observations of minimum, maximum, and modal prices across multiple markets, along with metadata on crop varieties and grades. The dataset contains over **24,000 records across 45 markets**, providing a comprehensive view of price behavior and volatility patterns.

All datasets were preprocessed through cleaning, outlier handling, and temporal alignment. Satellite-derived features, weather variables, and market data were synchronized to enable integrated modeling. This unified data pipeline supports crop classification, weather forecasting, price prediction, and subsequent economic analysis within a consistent framework.

2.2 Pipeline Description

The proposed framework is structured into three interconnected pipelines that collectively address crop identification, price dynamics, and decision support for agricultural markets:

(1) Crop Classification Pipeline: This component focuses on identifying crop types using multi-temporal satellite imagery and supervised machine learning. Sentinel-1 (SAR) and Sentinel-2 (optical) data are processed to extract spectral, temporal, and structural features, including vegetation indices, temporal growth patterns, and radar backscatter characteristics. Ground truth samples collected via Google Earth and Google Street View are used to train classification models such as Random Forest. The improved model incorporates multi-seasonal feature engineering and balanced sampling strategies to enhance generalization across regions, with a specific focus on accurate identification of onion crops in Nashik.

(2) Price Forecasting Pipeline: This module models and predicts onion price trends using historical mandi data. A Long Short-Term Memory (LSTM)-based neural network is

trained on time-series price data (2020–2025) to capture temporal dependencies, seasonality, and volatility patterns. The model leverages preprocessed modal price sequences and generates short- to medium-term forecasts, enabling anticipation of price fluctuations in agricultural markets.

(3) Price Stabilization and Market Analysis Pipeline: This component focuses on understanding and mitigating price volatility through data-driven analysis. Extensive exploratory data analysis (EDA) is performed on Nashik mandi data to identify seasonal trends, peak price periods, and inter-annual variability. The analysis evaluates existing government interventions such as import–export controls and proposes an alternative storage-based stabilization strategy. By enabling farmers to store produce post-harvest and sell during high-price periods, this approach aims to reduce distress selling and improve income stability.

Additionally, weather variables are analyzed in conjunction with price data to understand their influence on market behavior. Correlation analysis and temporal comparisons between climatic factors and prices provide insights into how environmental conditions indirectly affect supply and price fluctuations.

2.2.1 Crop Classification Pipeline

2.2.2 1.Crop classification

Data Preparation and Study Area Definition

The crop classification pipeline was implemented using **Google Earth Engine (GEE)** for large-scale geospatial processing. The study area was defined as the state of **Uttar Pradesh, India**, extracted from the FAO GAUL administrative boundary dataset. All satellite data were spatially clipped to this Area of Interest (AOI).

Multi-temporal Sentinel-2 Image Collection

Multi-temporal **Sentinel-2 surface reflectance imagery** was used to capture crop phenological variations across agricultural seasons. Median composites were generated for multiple time windows between **July 2022 and November 2023**, while filtering scenes with less than 10% cloud cover to minimize atmospheric contamination.

Each composite image includes spectral bands

$$\{B2, B3, B4, B5, B6, B8, B9, B11, B12\},$$

covering visible, red-edge, near-infrared, and shortwave infrared regions. All images were radiometrically scaled and temporally tagged to preserve seasonal information.

Vegetation Index Computation

To enhance crop separability, vegetation indices were derived from each temporal image. The **Normalized Difference Vegetation Index (NDVI)** and **Soil Adjusted Vegetation Index (SAVI)** were computed as:

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (2.1)$$

$$SAVI = \left(\frac{NIR - Red}{NIR + Red + L} \right) (1 + L), \quad L = 0.5 \quad (2.2)$$

where $NIR = B8$ and $Red = B4$. These indices capture vegetation vigor and soil-adjusted canopy response, which are critical for distinguishing crop types with similar spectral characteristics.

All spectral bands and vegetation indices across multiple periods were stacked into a single multi-band feature image. **2. Onion classification**

Ground Truth Integration

Ground truth crop samples were manually collected using **Google Earth and Google Earth Street View**. Crop types were visually verified at specific geographic locations, and corresponding latitude, longitude, and crop class labels were recorded in a structured CSV file. Additional samples representing non-crop classes such as water bodies and barren land were included to improve classification robustness.

These point-based ground truth samples were overlaid onto the multi-band feature image.

x	y	time-stamp	class	
27.96773	77.76305	2022-9	0	1
27.96677	77.76204	2022-9	0	
27.96604	77.7613	2022-9	0	
27.96562	77.76084	2022-9	3	
27.96546	77.75989	2022-9	3	
27.96582	77.7601	2022-9	2	
27.96579	77.75946	2022-9	4	
27.96299	77.76077	2022-9	5	
27.91684	77.62298	2022-9	3	
27.91625	77.62339	2022-9	8	
27.91472	77.6256	2022-9	7	
27.91322	77.62784	2022-9	7	
27.91179	77.63036	2022-9	7	
27.90796	77.64363	2022-9	5	

Figure 2.1: GT Data example values

Supervised Classification

Supervised classification was performed using **Random Forest (RF)** and **Support Vector Machine (SVM)** classifiers. Random Forest was implemented with 100 decision trees due to its robustness to overfitting and effectiveness in handling high-dimensional feature spaces.

Given a feature vector $\mathbf{x} \in R^d$, the Random Forest classifier predicts the class label y as:

$$y = \text{mode}\{h_k(\mathbf{x})\}_{k=1}^K \quad (2.3)$$

where h_k represents the k -th decision tree and K is the total number of trees. The dataset was split into 70% training and 30% testing samples. Performance was evaluated using confusion matrices, overall accuracy, and Cohen’s kappa coefficient. The output of this pipeline is a spatially explicit crop classification map covering approximately **15 crop classes**.

2.2.3 2. Onion Crop Classification Pipeline (Nashik)

Study Area and Implementation Platform

The onion-specific crop classification pipeline was implemented using **Google Earth Engine (GEE)** for scalable geospatial computation. The study area was defined as the **Nashik region in Maharashtra, India**, one of the largest onion-producing zones. The Area of Interest (AOI) was specified as a rectangular bounding region and all satellite datasets were spatially clipped accordingly.

Multi-season Data and Temporal Setup

To capture crop growth dynamics, a **multi-seasonal framework** was adopted spanning six agricultural cycles from **October 2020 to March 2026**. Each season consists of six monthly observations (October to March), corresponding to the onion growing cycle. Ground truth data were available for three seasons (2022–23, 2023–24, and 2025–26), while the remaining seasons were used for inference.

Satellite Data and Preprocessing

Multi-temporal **Sentinel-2 surface reflectance imagery** was used as the primary data source. Cloud masking was performed using QA60 band filtering, followed by radiometric scaling. Monthly median composites were generated with a cloud threshold of 20% to ensure data quality.

In addition, **Sentinel-1 SAR data** (VV and VH polarizations) were incorporated to capture structural and moisture-related crop characteristics, especially under cloudy conditions. Radar-derived features such as backscatter intensity and VH/VV ratio were also computed.

Feature Engineering

A comprehensive set of features was constructed to capture spectral, temporal, and phenological characteristics:

- **Vegetation and spectral indices:** NDVI, NDRE, GCVI, Moisture Index (MOIS), Bare Soil Index (BSI), NDTI, and chlorophyll gap.
- **Temporal features:** Monthly values across six time steps and inter-month differences (Δ features) to capture growth trends.
- **Ratio features:** Cross-index ratios (e.g., NDVI/BSI, NDRE/MOIS) to improve separability between crop and non-crop classes.
- **Phenological features:** Seasonal statistics such as mean and maximum NDVI and NDRE.
- **Spatial features:** Latitude information to capture geographic variation.
- **SAR features:** Seasonal mean VH backscatter, VH/VV ratio, peak-season VH response, and spatial texture.

All features were stacked into a high-dimensional multi-band image representing the full crop growth cycle.

Ground Truth and Sampling Strategy

Ground truth data were manually collected using **Google Earth Street View**, consisting of **320 labeled points** for onion and non-onion classes. To address class imbalance, a balanced sampling strategy was applied by limiting both classes to equal sample sizes before training.

Model Training

A **Random Forest classifier** was trained using selected high-importance features derived from feature importance analysis. The model was configured with:

- Number of trees: 150

- Minimum leaf population: 15
- Bagging fraction: 0.5

The classifier was trained in probabilistic mode, outputting class probabilities for each pixel.

Classification and Post-processing

For each season, probability maps were generated and converted into binary onion masks using a threshold of **0.46**. Spatial smoothing was applied using focal filters to reduce noise and improve spatial coherence of predictions.

Model Evaluation

Model performance was evaluated using both:

- **Leave-One-Season-Out (LOSO) cross-validation**, where the model was trained on two seasons and tested on the third.
- **In-sample validation** using training seasons.

Evaluation metrics included overall accuracy and Cohen’s kappa coefficient. Additionally, misclassified samples were analyzed and visualized to identify false positives and false negatives, enabling further model refinement.

Output Generation

Final outputs consist of season-wise onion crop masks at **10 m spatial resolution**, exported for further analysis. These outputs enable temporal tracking of onion cultivation patterns and serve as input for downstream price and market analysis pipelines.

2.2.4 2.LSTM-Based price forecasting of onion prices

Data Description

The price forecasting model was developed using historical **onion mandi price data** from the Nashik region, covering the period from **2020 to 2025**. The dataset consists of daily observations of *minimum*, *maximum*, and *modal prices*, along with market-level metadata. Among these, the **modal price** was selected as the primary target variable, as it best represents the prevailing market price.

Data Preprocessing

The raw time-series data were preprocessed to ensure consistency and suitability for deep learning models. Missing values were handled through interpolation, and outliers were removed based on statistical thresholds. The time series was then resampled (if required) to maintain uniform temporal intervals.

To stabilize training and improve convergence, the price values were **normalized using Min-Max scaling**:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where x represents the original price and x' is the scaled value.

Exploratory Data Analysis

Extensive exploratory data analysis (EDA) was conducted to understand price behavior. Seasonal trends, yearly variations, and volatility patterns were examined using time-series plots, rolling averages, and distribution analysis. The analysis revealed strong **seasonality** in onion prices, with recurring peaks during specific months and significant inter-annual fluctuations.

Sequence Generation

To model temporal dependencies, the time series was transformed into supervised learning format using a sliding window approach. Input sequences of fixed length T (look-back window) were used to predict the next time step:

$$\{x_{t-T}, x_{t-T+1}, \dots, x_{t-1}\} \rightarrow x_t$$

This formulation enables the model to learn both short-term and long-term dependencies in price movements.

LSTM Model Architecture

A **Long Short-Term Memory (LSTM)** network was employed due to its effectiveness in modeling sequential data and capturing temporal dependencies. The architecture consists of:

- One or more LSTM layers to capture temporal patterns
- Dropout layers for regularization and prevention of overfitting
- Fully connected (Dense) output layer for price prediction

The model was trained using the **Adam optimizer** with Mean Squared Error (MSE) as the loss function.

Model Training and Evaluation

The dataset was split into training and testing sets based on temporal order to avoid data leakage. The model was trained over multiple epochs with batch processing.

Performance was evaluated using standard regression metrics such as:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Predicted values were inverse-transformed to the original price scale for interpretation.

Results and Forecasting

The trained LSTM model successfully captured seasonal patterns and price trends in the data. The model was used to generate **future price forecasts**, enabling anticipation of price spikes and declines.

Table 2.1: Performance Metrics of LSTM-Based Onion Price Forecasting Model

Metric	Value
MAE	124.22
RMSE	164.45
R^2 Score	0.3452
MAPE	0.1123
Accuracy (%)	88.77

Integration with Analysis Pipeline

The predicted price trends were further analyzed alongside weather variables and market behavior to understand underlying drivers of price fluctuations. These forecasts also support the proposed price stabilization strategy by identifying optimal selling periods for farmers.

2.2.5 3.Price stabilization study

Motivation

Agricultural markets in India often experience significant price volatility due to seasonal supply fluctuations, demand uncertainty, and limited storage infrastructure. Perishable crops such as onion are particularly vulnerable, where prices typically drop sharply during harvest periods due to excess supply and rise later due to scarcity. To manage such fluctuations, government interventions such as Minimum Support Price (MSP), export bans, and import policies are frequently employed. However, these approaches are often reactive, costly, and may distort market dynamics.

Proposed Approach

This study explores an alternative, data-driven price stabilization strategy based on **decentralized storage systems**. The key idea is to enable farmers to store their produce during periods of excess supply and sell it during high-price periods, thereby smoothing price fluctuations and improving income stability.

The feasibility of this approach relies on understanding the underlying **demand–supply dynamics**, which can be inferred from historical price trends. By analyzing price variations over time, it is possible to identify recurring seasonal patterns, peak demand periods, and price recovery phases following harvest cycles.

Exploratory Analysis Framework

To capture these dynamics, extensive **exploratory data analysis (EDA)** was conducted on onion mandi price data from Nashik (2020–2025). The analysis focused on:

- Temporal price trends (daily, monthly, and yearly)
- Identification of seasonal peaks and troughs
- Volatility patterns and inter-annual variability
- Correlation between price and weather variables (temperature, humidity, soil moisture)

The price vs time representation serves as a key tool to visualize supply shocks and recovery phases. A typical seasonal trend is illustrated in Figure 2.2, where prices decline during harvest months and gradually increase as stored supply reduces in the market.

Interpretation

From such temporal patterns, it is possible to approximate periods of **oversupply (low prices)** and **undersupply (high prices)**. This information can be leveraged to guide storage decisions. Instead of immediate selling during low-price periods, farmers can delay market entry until favorable price conditions emerge.

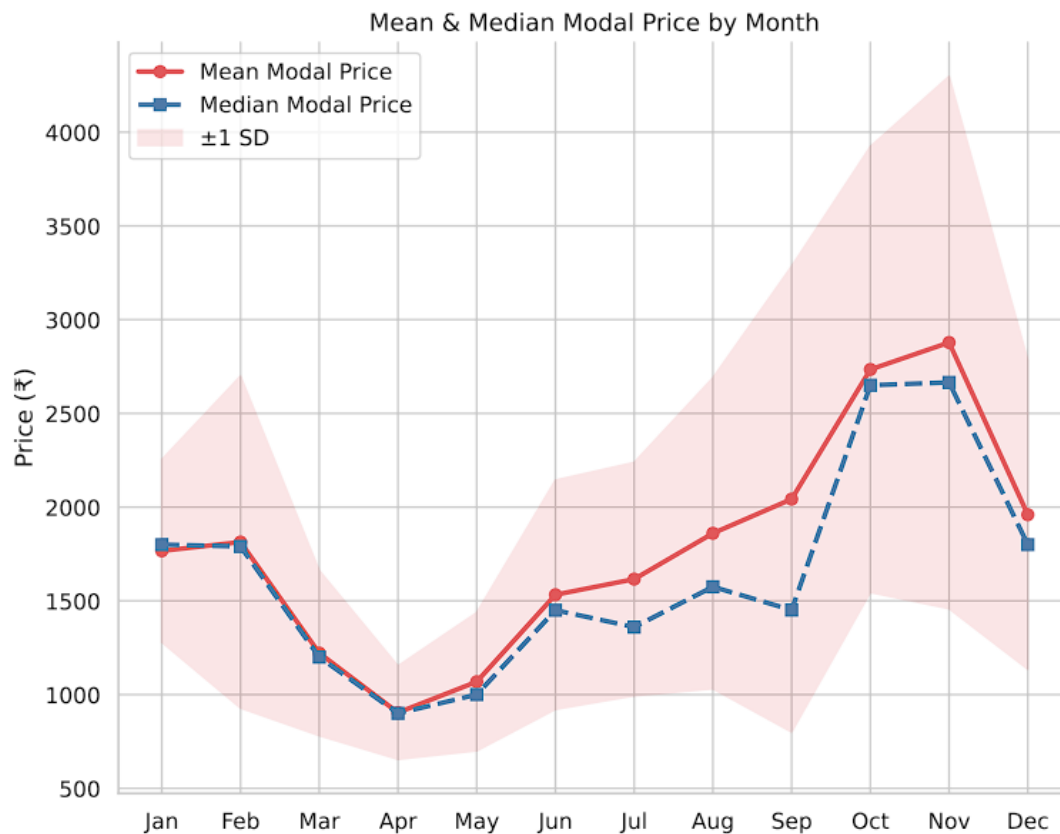


Figure 2.2: Illustrative onion price vs time trend showing seasonal fluctuations

Implications

By combining price trend analysis with basic storage infrastructure, the need for heavy government intervention can be reduced. A low-cost, accessible storage system allows farmers to actively participate in price optimization rather than being constrained by market timing. This approach shifts the system from a reactive policy-driven model to a **proactive, data-informed decision framework**, improving both market stability and farmer welfare.

Further quantitative evaluation of this strategy is presented in the Results section.

3. Results

3.0.1 Classification results

1.Crop classifier

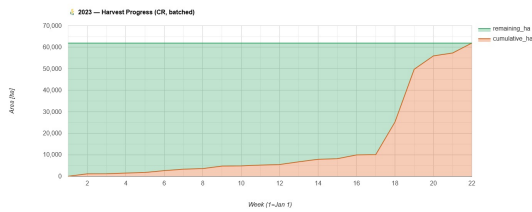
Table 3.1: Performance Evaluation of Random Forest Classifier

Metric	Training Data	Testing Data
Overall Accuracy	0.9989	0.9754
Cohen's Kappa Coefficient	0.9942	0.8749
Confusion Matrix Size	18 elements	18 elements

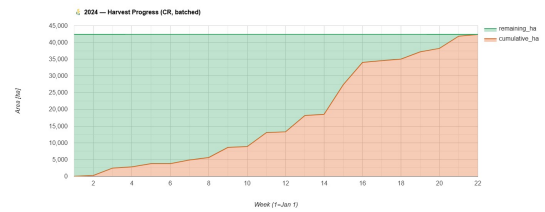
2.Onion classification

Table 3.2: LOSO Cross-Validation Results for Onion Classification

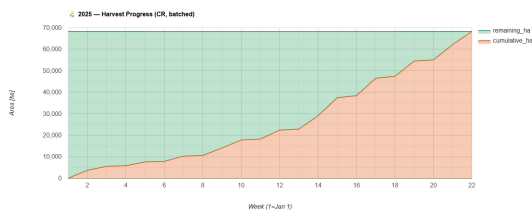
Test Season	Confusion Matrix	OA	κ
Oct 2022–Mar 2023	$\begin{bmatrix} 29 & 13 \\ 10 & 55 \end{bmatrix}$	0.7850	0.5435
Oct 2023–Mar 2024	$\begin{bmatrix} 39 & 24 \\ 12 & 46 \end{bmatrix}$	0.7025	0.4088
Oct 2025–Mar 2026	$\begin{bmatrix} 21 & 11 \\ 3 & 37 \end{bmatrix}$	0.8056	0.5962



(a) 2023



(b) 2024



(c) 2025

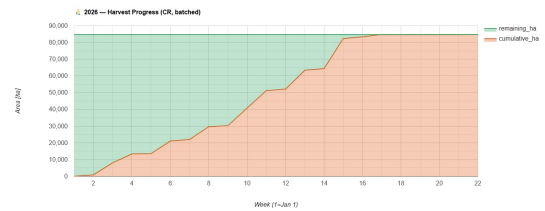
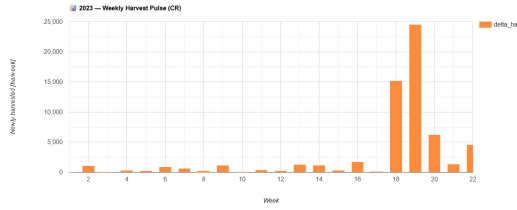
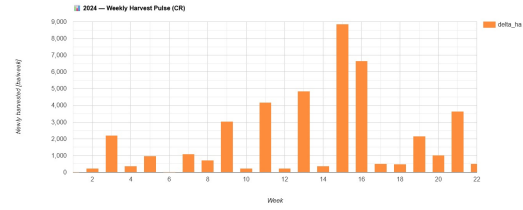


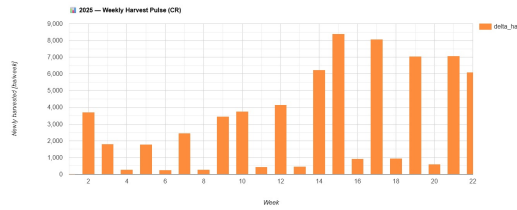
Figure 3.1: Harvest Progress of Onion Crop Across Different Years



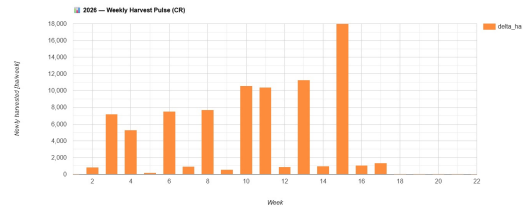
(a) 2023



(b) 2024

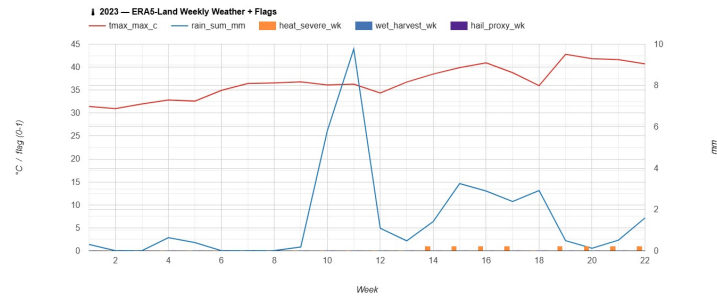


(c) 2025

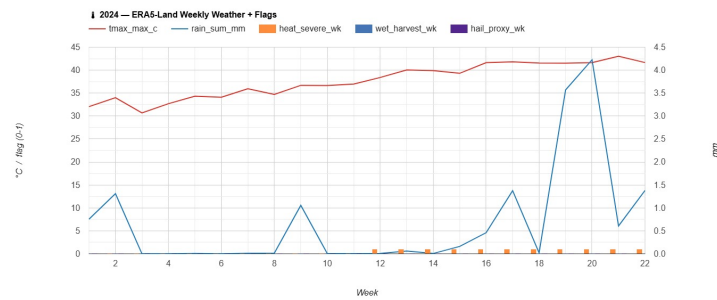


(d) 2026

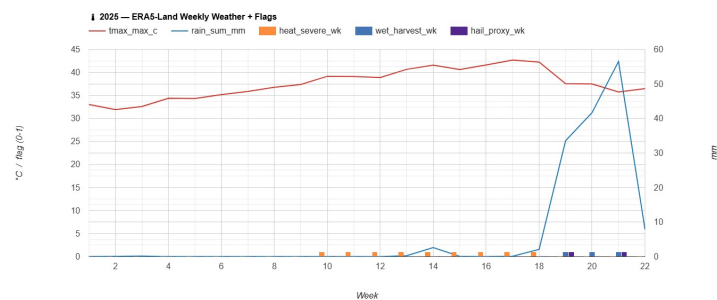
Figure 3.2: Weekly Harvest Pulse (Newly Harvested Area) for Onion Crop Across Different Years



(a) 2023



(b) 2024



(c) 2025

Figure 3.3: Weekly Weather Trends and Event Flags (Temperature, Rainfall, and Harvest Conditions) Across Years

3.0.2 LSTM Price fore-casting of onion

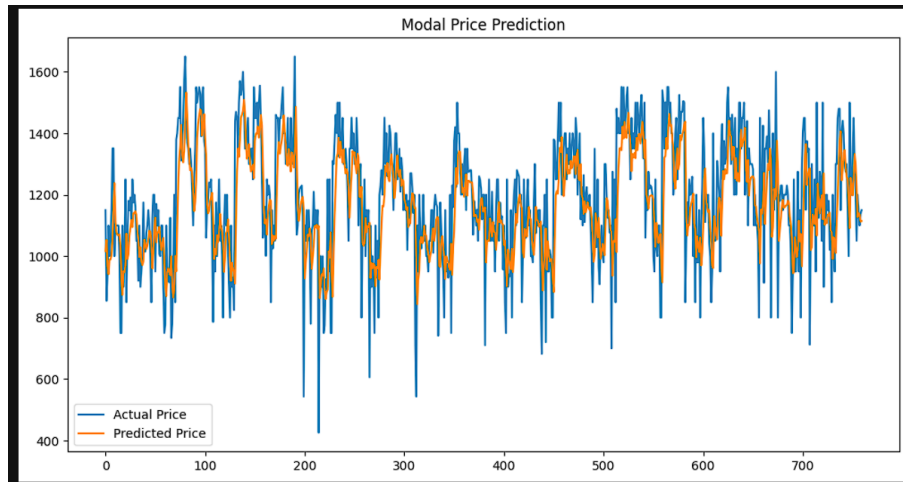


Figure 3.4: Price prediction(yellow-prediction,blue-actual)

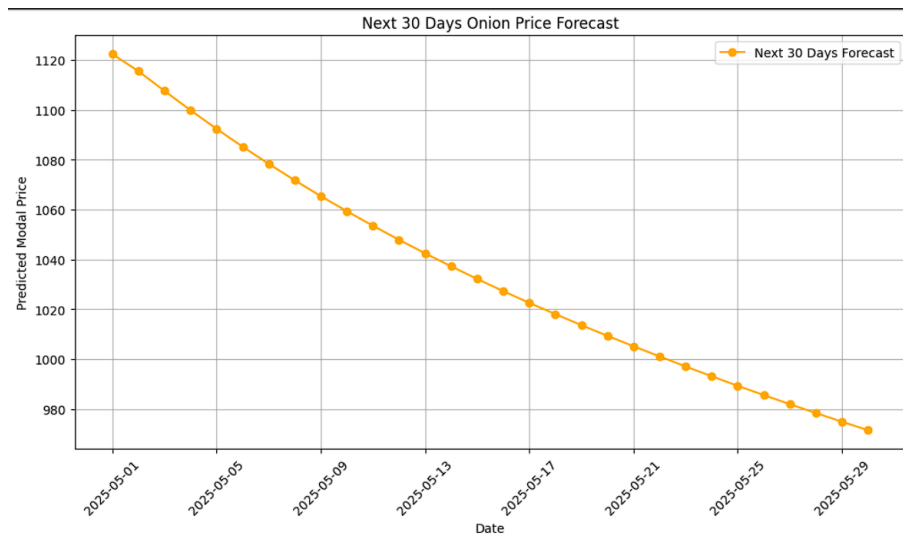


Figure 3.5: 30 days forecast

Table 3.3: Performance Metrics of LSTM-Based Onion Price Forecasting Model

Metric	Value
MAE	124.22
RMSE	164.45
R^2 Score	0.3452
MAPE	0.1123
Accuracy (%)	88.77

3.0.3 Price stabilisation study using analysis

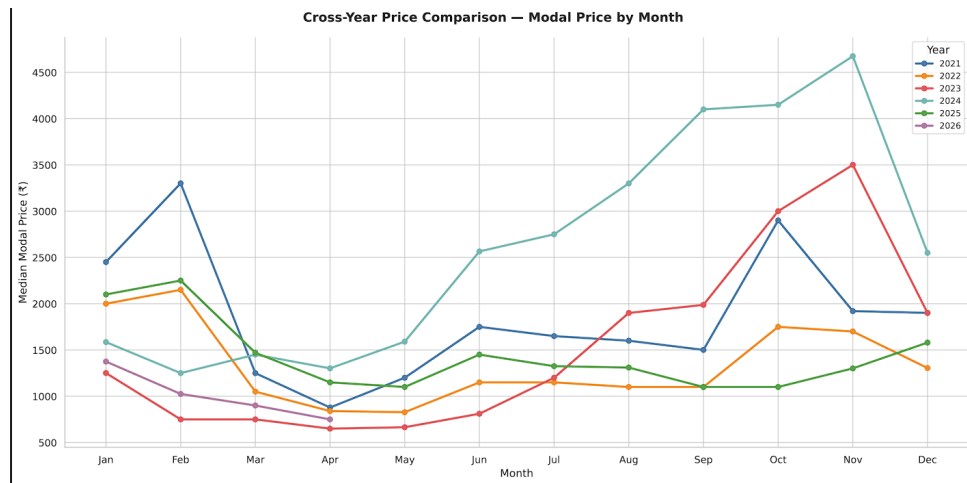


Figure 3.6: MSP price variation across years 2021-2026

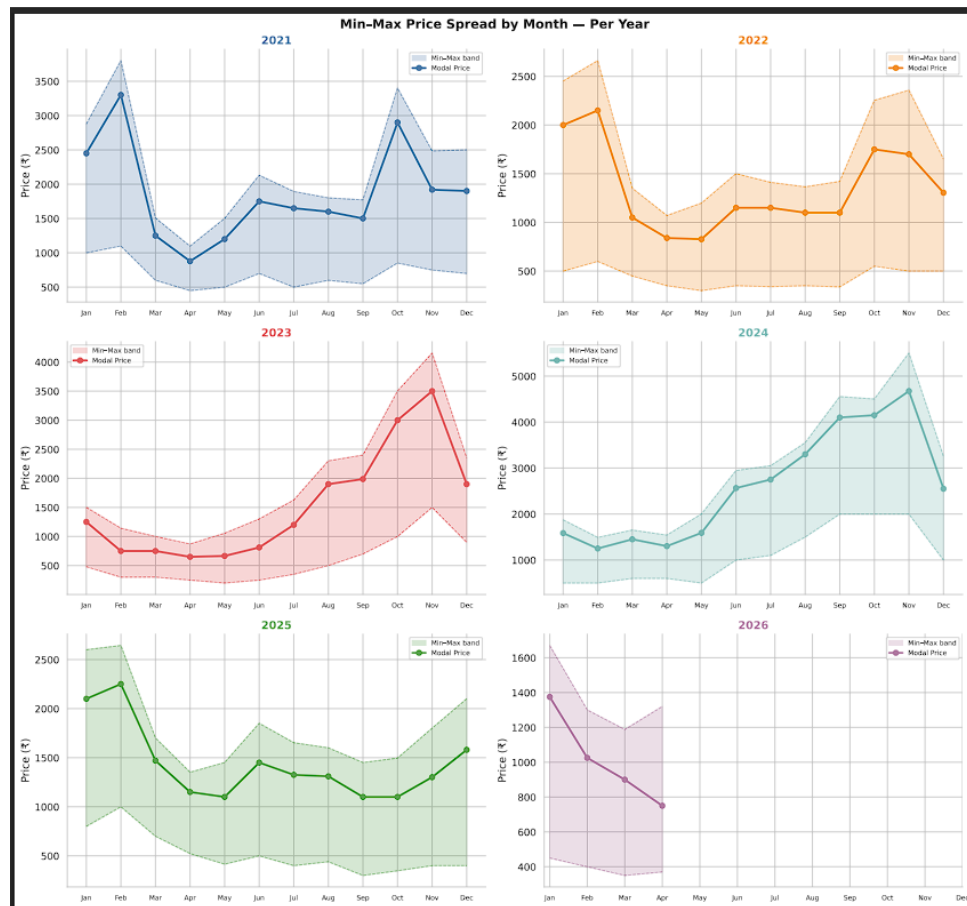


Figure 3.7: Min max and modal price distribution

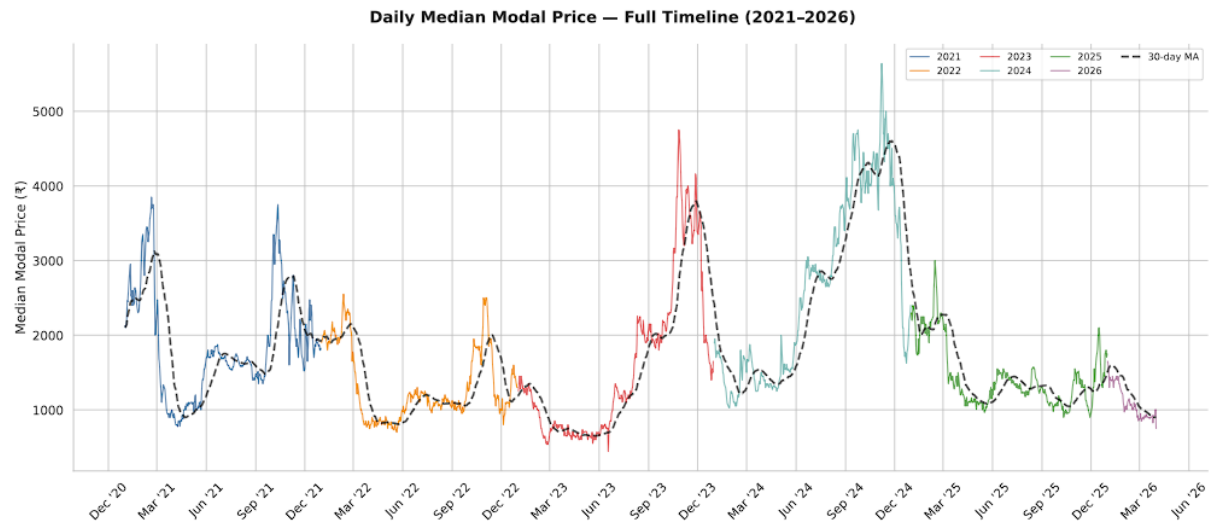


Figure 3.8: Daily median modal price distribution

4. Conclusion/Future Work

We did mandi study and found out that prices of cereals more or less remained same throughout the year and these crops also are resistance to fluctuation in climate. But Top 3 consumed vegetables of India Onion, Tomato and Potato have high fluctuations. Geography wise we see negligible differences in rate signifying good and cheap transportation. But for the big 3 vegetable if we keep the month same and look at data for different years there is a huge fluctuation in rate for all 3 vegetables which are caused by Extreme weather events like Unseasonal rains, heatwaves, droughts, floods. So in future we would like to focus on finding the underlying patterns affecting these rates using our Classifier model on satellite data to see where are the clusters of these 3 vegetables are grown the most, and use weather data to see how that relates to price fluctuation.

Bibliography